Routing a maximal shared crossing

Human Compression project: L044 continuation

Throughout, $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, is transitive, has a connected string C-marking, and every maximal omission G_k is intransitive. The alternating rank bound is assumed at strictly smaller degrees at least twelve. All previously accepted bounded components, including the complete both-imprimitive bound and the finite small-tail inputs, remain paid. An eligible k-edge crosses two G_k-orbits. A unique eligible edge is a bridge of the full permutation graph and joins its two omission orbits.

Lemma 1 (Interval choice in the contradiction range)**.** *Suppose $r > \lfloor (n-1)/2 \rfloor$, and a unique eligible k-edge has two nontrivial omission orbits. At least one of their active index sets is a single interval. If a label larger than k is shared by both orbits, that interval lies to the right of k; if every shared label is smaller than k, it lies to the left.*

Proof. If both orbit actions are imprimitive, the complete accepted both-imprimitive theorem gives the excluded rank bound. Hence at least one action is primitive. The accepted primitive-orbit interval lemma gives a single active interval in degree at least three. Its complete two-point exception theorem supplies the same conclusion in degree two under the present high-rank hypothesis. This uses faithful commuting factors and the complete L047 theorem, not IP in an orbit projection. An active interval avoids the omitted index k. A shared label belongs to both active sets, so its side determines the side of this interval. In the last assertion a shared label is understood to exist, as in its applications below. □

Theorem 2 (A complete numerical maximal-crossing alternative)**.** *Suppose some unique eligible crossing has a shared label to its right. Choose its index i maximal among all such crossings. Then at least one of the following holds:*

1. *$r \le \lfloor (n-1)/2 \rfloor$;*
2. *an actual first or last three-generator subgroup is one of the two six-point A_5 markings of source Table 2, allowing full string reversal;*
3. *G_{r-1} has a singleton orbit.*

The alternatives refer to actual supported subgroups and actual omission orbits, not projected IP or unnamed isomorphisms.

Proof. Assume the rank bound fails and that no exceptional actual end triple occurs. We prove the singleton conclusion. Let the unique i-edge be ab, joining omission orbits A, B, and let $h > i$ be the last shared label. Both orbits are nontrivial. For $i > 0$, Lemma 1 lets us choose B with its active interval to the right, so $G_{<i}$ fixes B pointwise. The accepted transitive-side and completed intransitive-side support alternatives, according to the action of $G_{<i}$ on A, give

$$h < r - 1, \quad X = \operatorname{supp}(G_{>h}) \subseteq B \setminus \{b\}, \quad 2h \le n - |X| - 1. \tag{1}$$

For $i = 0$, the accepted endpoint shared-label exclusion gives $h < r - 1$. All later labels act on just one of A, B; they must act on the same side, since a change of side makes adjacent generators commute, disconnecting the string diagram. Name that side B. The complete endpoint fixed-support bound gives exactly (1). Thus (1) holds in both cases.

Put $R = G_{>h}$. If it is transitive on X, the complete accepted transitive-tail alternative contradicts our two exclusions. Consequently R is intransitive on its support. If every colour larger than h had two ambient-eligible edges, the accepted ambient terminal theorem would give

$$2(r-1-h) \le |X| - q \le |X| - 2,$$

where $q \ge 2$ counts its nontrivial orbits. Together with (1) this gives $2r \le n-1$, again a contradiction. Every colour has at least one eligible edge, by transitivity of G and intransitivity of its omission. Hence there is a smallest $j > h$ with a unique eligible edge, say cd.

Since ρ_j is supported on $X \subset B$, its bridge is inside B. Delete the two bridges. The resulting connected regions are A, M, D, with $a \in A$, $b, c \in M$, $d \in D$; the G_j-orbit containing the i-edge is $C = A \cup M$. In particular C is nontrivial. If D is a singleton and $j < r-1$, the complete interior singleton-orbit bound contradicts rank failure. If it is a singleton and $j = r-1$, we have the desired conclusion. We may therefore assume D nontrivial.

Suppose no label is shared across the j-cut. Each generator other than ρ_j then acts on exactly one of C, D. Along either string interval $< j$ or $> j$ the side cannot change, because adjacent noncommuting generators cannot have disjoint supports. The prefix contains the i-edge and hence acts on C. Nontriviality of D forces $j < r-1$ and makes the suffix act on D. Thus these are actual supported prefix and suffix groups, transitive on their respective orbits. The complete supported-interval alternative contradicts our two exclusions. Therefore a shared label exists; let g be the first. By maximality of i, no label larger than j is shared at this crossing. Thus $g < j$.

We record that ρ_0 acts nontrivially on C. For $i = 0$ its crossing edge lies in C. For $i > 0$, it is a nonidentity generator of $G_{<i}$, supported on $A \subset C$ by our orientation. This observation fixes the second cut's orientation as follows.

If $j = r-1$, the dual endpoint shared-label exclusion rules out $g = 0$. All indices less than g are unshared and form one connected string interval. They consequently act on the same side, which is C because of ρ_0. The dual endpoint support bound then gives

$$Y = \operatorname{supp}(G_{<g}) \subseteq C, \qquad 2(r-1-g) \le n - |Y| - 1. \tag{2}$$

The suffix $G_{>j}$ is empty in this case.

Suppose instead $j < r-1$. By Lemma 1, one of C, D has an active interval to the left of j. If it were D, then $G_{>j}$ would be supported on C. Apply the complete interior support alternative in reversed index order, with first orbit C and second orbit D. Under rank failure it gives $g > 0$ and makes $G_{<g}$ supported on D. This is impossible because ρ_0 acts on C. Therefore the interval side is C, and $G_{>j}$ fixes C pointwise. Apply the same reversed support alternative with first orbit D and second orbit C. It gives $g > 0$ and (2). Both transitive and intransitive actions of this reversed prefix are covered by the accepted alternatives. We have proved the second orientation and budget, not assumed them.

Now the complete nested-bridge count applies: its outside regions A, D are nontrivial, $G_{<i}$ fixes $M \cup D$, and $G_{>j}$ fixes $A \cup M$. It gives $g \ge h > i$. Since g is the first shared label at j and ρ_i acts on C, this also proves $\rho_i|_D = 1$. Similarly $j > h$ and the last-shared definition at i prove $\rho_j|_A = 1$. These are the purity hypotheses needed for the complete interior and endpoint touching exclusions. Those exclusions therefore give $g > h$.

Set $P = G_{<g}$. If it is transitive on its support Y, the full transitive-tail alternative in the second cut's reversed orientation again contradicts the excluded exceptional ends. Thus P, R are both intransitive on their supports. All colours strictly between h and g have two ambient-eligible edges, because $g < j$ and j was the first unique crossing after h. Equations (1)–(2) and the complete orbit-incidence deficit theorem now give $2r \le n-1$. This final contradiction proves the theorem. □

Source, attribution and exact limit. This is a complete replacement of the numerical routing in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.28 (physical pages36–37). The source's maximal shared crossing and minimal subsequent crossing strategy is retained. The proper index reversal, degree-two interval exception, actual orientation, touching exclusions and orbit-incidence count are supplied by the paid preceding proofs. No new census or formal declaration is needed.

The literal source conclusion specifies the *last* exceptional triple; the theorem here retains either actual end, which is the alternative supplied by the complete transitive-tail theorem. It does not assert that this end-choice strengthening has been proved. The main alternating-group recovery still needs the exceptional-head and singleton continuation, full paired-square rigidity and the exact remaining source table/disposition assembly. No whole L044, original, question or priority credit is claimed.